

Feedback Control Systems (EE3004)**Sessional-I Exam**Date: February 25th 2025

Total Time (Hrs): 1

Course Instructor

Total Marks: 30

Dr. Omer Saleem & Ms. Tamania Javed

Total Questions: 3

22L-6234

Roll No

6A

Section

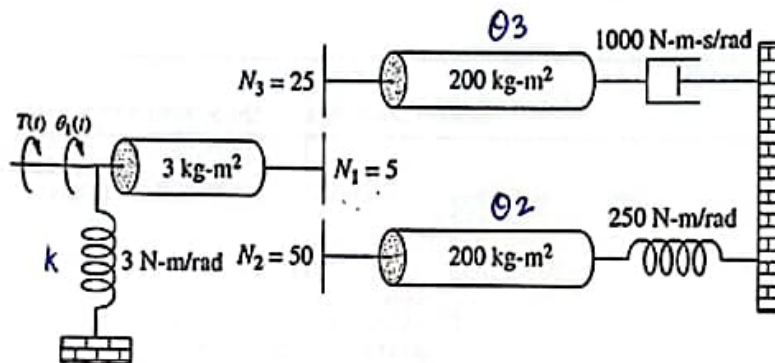
H. Saleem
Student Signature

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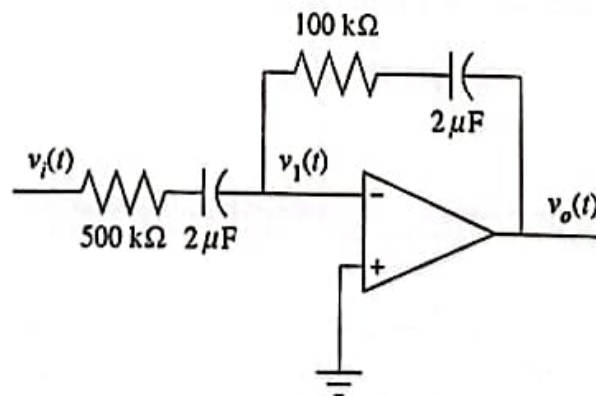
Attempt all the questions.

CLO # 1: Develop the mathematical model of a physical system

Q1: Formulate the transfer function, $G(s) = \theta_1(s) / T(s)$, for the rotational mechanical system shown in the Figure below. [10 marks]

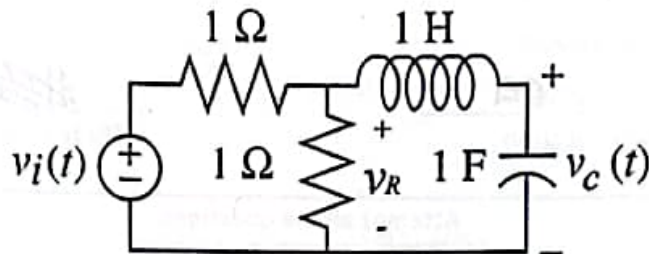
**CLO # 1: Develop the mathematical model of a physical system**

Q2: Formulate the transfer function $G(s) = V_o(s) / V_i(s)$ of the following inverting-amplifier circuit. Hence, use this transfer function to evaluate $v_o(t)$, if a unit-step signal is applied at $v_i(t)$. [10 marks]



CLO # 1: Develop the mathematical model of a physical system

Q3: Assume that currents $i_1(t)$ and $i_2(t)$ are flowing in loop 1 and loop 2 of the following circuit, respectively. The input of the circuit is $v_i(t)$. The voltage across the capacitor is $v_C(t)$. The output of the circuit is $v_R(t)$. Develop the state-space model of the system, identifying the state variables as well as the matrices A , B , C , and D . [10 marks]



Feedback Control Systems (EE3004)

Final Exam

Date: May 29th 2025

Total Time: 3 Hours

Course Instructor

Total Marks: 100

Dr. Omer Saleem and Ms. Tamania Javaid

Total Questions: 05

Semester: SP-2025

Campus: Lahore

Dept: Electrical Engg.

Student Name

Roll No

Section

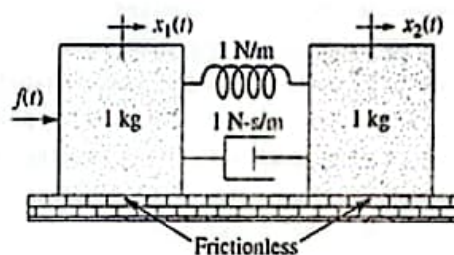
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CLO # 1: Develop the mathematical model of a physical system.**Q1:****[20 marks]**

- a. Develop the state space representation of the following mechanical system.



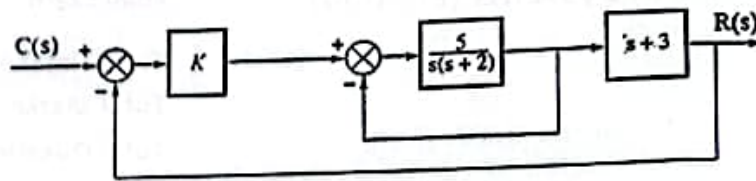
- b. Formulate transfer function of the form, $Y(s)/U(s)$, using the following state space representation. Hence, find the expression of output, $y(t)$, if the input signal, $u(t)$, is a unit-step signal.

$$\dot{x}(t) = \begin{bmatrix} -5 & 0 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 7 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

CLO # 2: Analyze the transient and steady state response of a feedback system.**Q2:****[20 marks]**

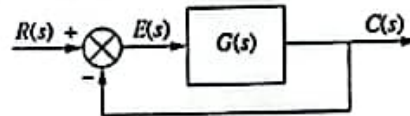
- a. Analyze the closed-loop system shown below.
- Reduce the system's block diagram to find its overall transfer function $T(s)$.
 - Find the value of K in the system to yield a natural frequency of 7.07 rad/s.
 - Hence, calculate the system's damping ratio as well as its settling time.



b. For the unity feedback system shown below, where:

$$G(s) = \frac{90}{s^n(s+m)}$$

- Find the values of n and m to yield a constant steady-state error of 0.01 for a unit-ramp input.
- Hence, find the value of n that would eliminate the steady-state error.



CLO # 3: Analyze the stability of a system.

Q3:

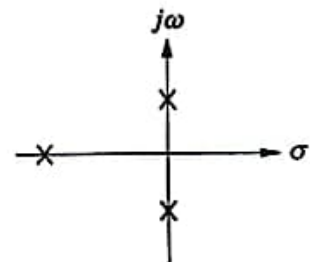
[20 marks]

a. For the unity feedback system shown below, where:

$$G(s) = \frac{K(s+4)}{s(s+1.2)(s+2)}$$

- Find the closed-loop transfer function, $T(s)$, of the system.
 - Hence, analyze $T(s)$ to calculate the range of K for stability.
- b. Use Routh Hurwitz criteria to calculate K that will place the closed-loop poles of the system as shown in the Figure here; one pole on real-axis and two poles on $j\omega$ -axis. The system's closed loop transfer function $T(s)$ is expressed below. Hence, find the value of the poles as well.

$$T(s) = \frac{2s^4 + (K+2)s^3 + Ks^2}{s^3 + s^2 + 2s + K}$$

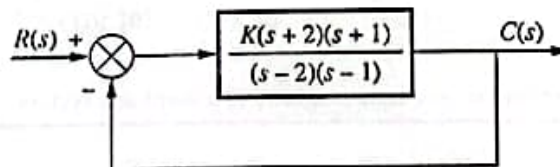


CLO # 4: Analyze a given system in frequency domain.

Q4:

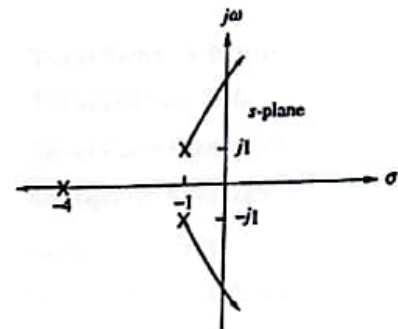
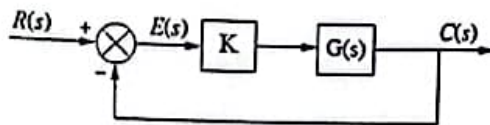
[20 marks]

a) For the system shown in Figure below,



- Find the break-away and break-in points (if any) of the root-locus.
- Find the crossings on the $j\omega$ -axis of the root-locus.
- Hence, illustrate the diagram of the root-locus.

- b) The root locus of a system, $G(s)$, is shown here:
- Find the open-loop transfer function, $G(s)$, of the system.
 - Hence, using the appropriate property, find the gain K for which the system will yield a closed-loop pole on the real-axis at $s = -5$.



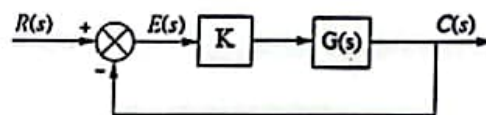
CLO # 5: Design a closed-loop feedback controller

Q5:

[20 marks]

The open loop transfer function of an un-compensated unity feedback system is given below.

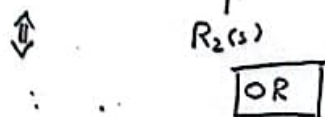
$$K G(s) = \frac{K}{(s+2)(s+6)}$$



The dominant pole of the un-compensated system occurs at $K = 29$.

- Find the location of the dominant pole of the un-compensated system.
- Find the un-compensated system's settling-time, peak-time, and percentage overshoot.
- It is desired to compensate the system to decrease its settling-time by a factor of 1.5 without affecting its overshoot. Find the new settling-time, peak-time, and the new dominant pole location to satisfy the required performance metrics.
- Justify which controller (PI or PD) is best suited for this application.
- Hence, design the controller selected in (d) to satisfy the desired performance metrics. Write the transfer function of the controller including its pole/zeros.

6) $R_1(s) \rightarrow \oplus \rightarrow G(s) \rightarrow C(s) = G(s)(R_1(s) + R_2(s))$
 $= G(s)R_1(s) + G(s)R_2(s)$



$R_1(s) \rightarrow G_1(s) \rightarrow \oplus \rightarrow C(s) = R_1(s)G_1(s) + R_2(s)G_2(s)$
 $= G(s)(R_1(s) + R_2(s))$

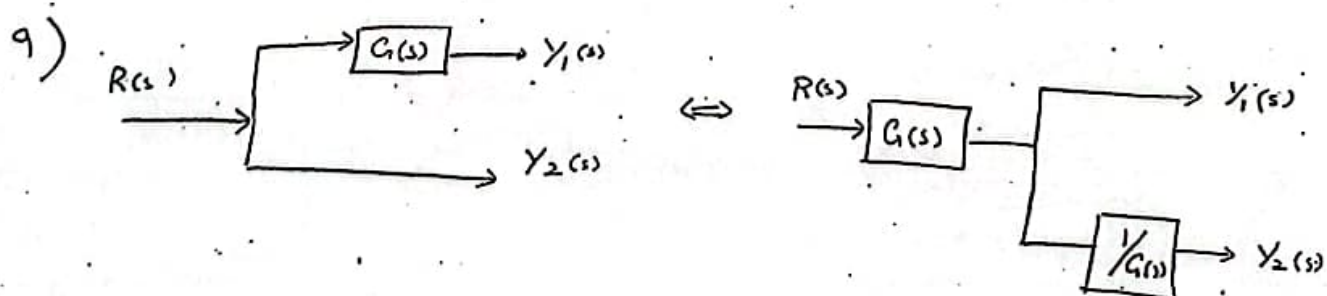
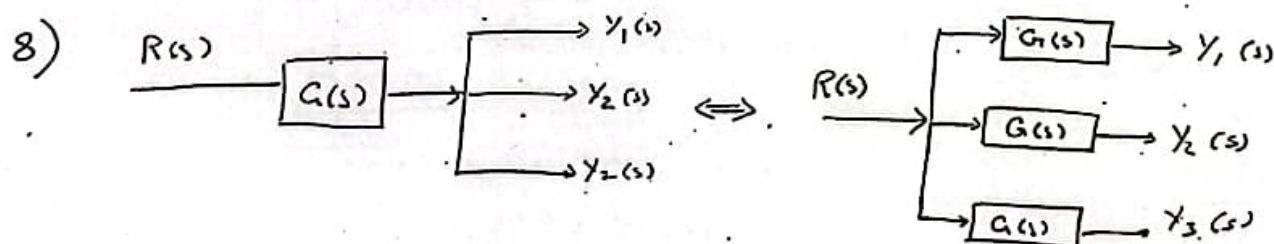
SAME as above

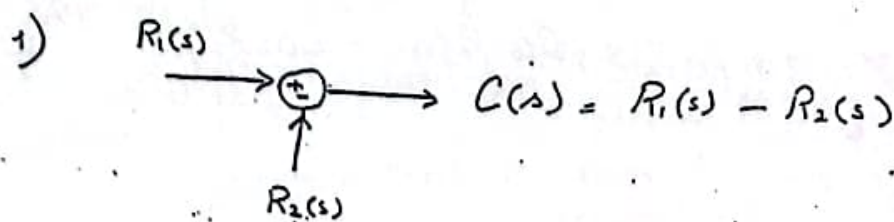
7) $R_1(s) \rightarrow G(s) \rightarrow \oplus \rightarrow C(s) = R_1(s)G(s) + R_2(s)$



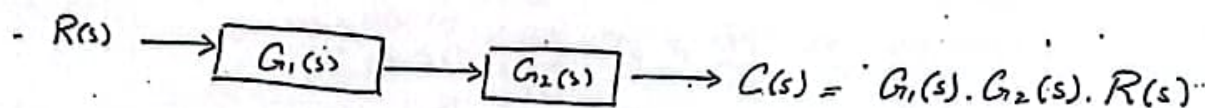
$R_1(s) \rightarrow \oplus \rightarrow G(s) \rightarrow C(s) = (R_1(s) + \frac{R_2(s)}{G(s)})G(s)$
 $= R_1(s)G(s) + R_2(s)$

SAME as above!

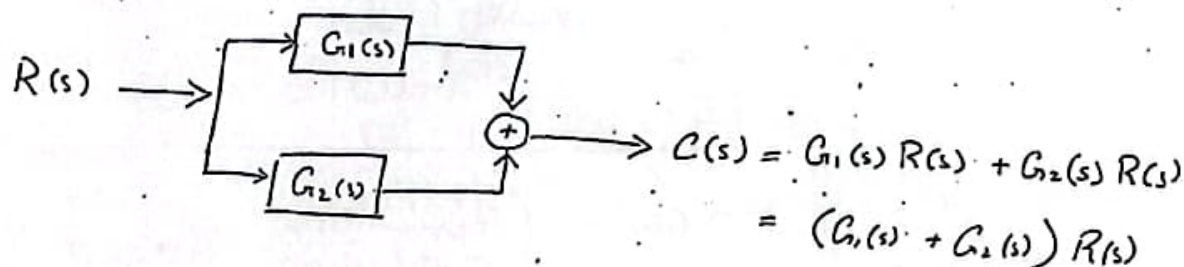




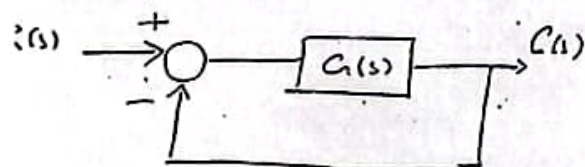
2) Series



i) Parallel

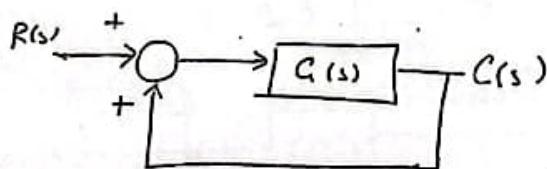


i) Unity Feedback



ve feedback

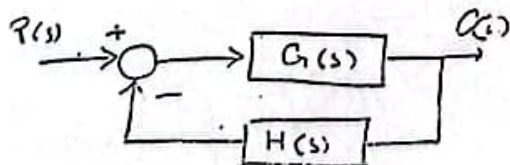
$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$



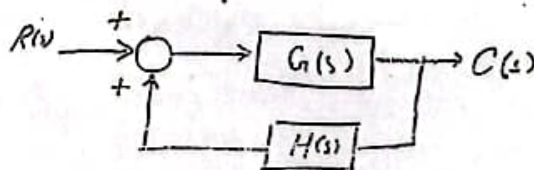
+ve feedback

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)}$$

Non-unity Feedback



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + H(s) G(s)}$$



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - H(s) G(s)}$$

Electrical System $\left[\text{Sum of imp in Mesh } \Delta \right] I_1(s) - \left[\text{Sum of imp common } \Delta, 2 \right] I_2(s) = \left[\text{sum of voltages in } \Delta \right]$

Mechanical System $\left[\text{sum of imp in } \Delta \right] y_1(s) - \left[\text{sum of imp common in } \Delta, 2 \right] y_2(s) = \left[\text{sum of applied forces} \right]$

Same for rotational system

Gears

$$\frac{T_1}{T_2} = \frac{N_1}{N_2}$$

$$\frac{\omega_1}{\omega_2} = \frac{N_2}{N_1}$$

$$J_{eq} = J \left(\frac{N_{dest}}{N_{sour}} \right)^2$$

$$D_{eq} = D \left(\frac{N_{dest}}{N_{sour}} \right)^2$$

$$k_{eq} = k \left(\frac{N_{dest}}{N_{sour}} \right)^2$$

State Space

Electrical

$$x = i_L$$

$$x = v_C$$

variable = No. of L and C

$$v_L = L \frac{di_L}{dt}$$

$$i_C = C \frac{dv_C}{dt}$$

$$AX + CU = \dot{x}$$

$$CX + DU = y$$

Mechanical

State = No. of Variable order

$$y_1 = x_1 ; y_2 = x_3$$

$$\frac{dy_1}{dt} = x_2 ; \frac{dy_2}{dt} = x_4$$

$$\frac{d^2 y_1}{dt^2} = \frac{dx_2}{dt} ; \frac{d^2 y_2}{dt^2} = \frac{dx_4}{dt}$$

$$\frac{dx_1}{dt} = x_2 ; \frac{dx_3}{dt} = x_4$$

BLOCK REDUCTION

Series $\rightarrow$ Multiply

Parallel $\rightarrow$ Add

$$\text{-ve feedback} \rightarrow \frac{G(s)}{1 + G(s)H(s)}$$

$$\text{+ve feedback} \rightarrow \frac{G(s)}{1 - G(s)H(s)}$$

$$\text{Unity} \rightarrow \frac{G_s}{1 + G_s}$$

Root Locus

① Sketch

② Asymptotes & centre

$$\sigma = \frac{180(2k+1)}{n-m}, k = 0, 1, 2, \dots$$

n = no of poles, m = zeroes

$$\sigma_a = \frac{\sum \text{pole} - \sum \text{zero}}{n-m}$$

③ Breaking point

$$1 + \frac{k}{(s+a)} = 0$$

$$k = -(s+a), \frac{dk}{ds} = 0$$

$$s = a, b \rightarrow k = a, b$$

④ $j\omega$ crossing

Routh table

-ve value $k \rightarrow$ No crossing

$$\phi_0 = 180^\circ - (\sum \text{pole } \phi)$$

$$+ \sum \text{zero } \phi$$

$$\phi_A = 180^\circ - (\sum \text{zero } \phi) + \sum \text{pole } \phi$$

Desired poles

$$= -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2}$$

$$180^\circ = \sum \text{pole } \phi + \sum \text{zero } \phi$$

$$\frac{Y(s)}{U(s)} = C(SI - A)^{-1} B + D$$

1st order system

$$c(t) = (1 - e^{-at})ut$$

Transient Condition

Stability

Routh table

Value of k put $k = 0$

sign change
Total-LHP=RHP
LHP $\rightarrow$ stable

Steady state Error

Type 0/step

$$G(s) = \frac{1}{(s+a)(s+b)}$$

$$e_{ss} = \frac{1}{1 + kp}$$

$$kp = \lim_{s \rightarrow 0} s G(s)$$

Type 1/Ramp

$$G(s) = \frac{1}{s(s+a)(s+b)}$$

$$e_{ss} = \frac{1}{kv}$$

$$kv = \lim_{s \rightarrow 0} s G(s)$$

$$ka = \lim_{s \rightarrow 0} s^2 G(s)$$

Value of k

$$k = \frac{1}{|G(s)H(s)|}$$

When

$$= \frac{1}{(s+a)(s+b)}$$

$$k = |G(s)H(s)|$$

When

$$= \frac{k}{(s+a)(s+b)}$$

$$G(s)H(s) = (s+a)(s+b)$$

Gain

$$PI = \frac{k(s+z)}{s}$$

$$PD = k(s+z)$$

$$\text{Lead} = \frac{k(s+z)}{(s+p)}$$

$$\text{Lag} = \frac{k(s+z)}{(s+p)}$$

steady state

PI, Lag

Transient

PD, Lead

2nd order

$$\xi = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$T_r = \frac{2.2}{\omega_n}$$

$$OS = e^{\frac{-\pi \zeta}{\sqrt{1 - \zeta^2}}} \times 100$$

$$\zeta = \frac{+\ln(OS)}{\sqrt{\pi^2 - \ln^2(OS)}}$$

ζ should be

$$0 < \zeta < 1$$

Type 2/parabolic

$$G(s) = \frac{1}{s^2(s+a)(s+b)}; e_{ss} = \frac{1}{ka}, ka = \lim_{s \rightarrow 0} s^2 G(s)$$

Function	Z_1	Z_2	Transfer Fn
proportional	$\text{---} \text{---} R_1$	$\text{---} \text{---} R_2$	$= -R_2/R_1$
integral	$\text{---} \text{---} R_1$	$\text{---} \text{---} C$	$= -1/RC$
Derivative	$\text{---} \text{---} \text{---}$	$\text{---} \text{---} \text{---}$	$-(RC)s$
PI	$\text{---} \text{---} R_1$	$\text{---} \text{---} \text{---}$	$-\frac{R_2}{R_1} \left(\frac{s + 1/RC}{s} \right)$
PD	$\text{---} \text{---} R_1$	$\text{---} \text{---} R_2$	$-R_2 C \left(s + \frac{1}{R_1 C} \right)$
PID	$\text{---} \text{---} \text{---}$	$\text{---} \text{---} \text{---}$	$-\left[\left(\frac{R_2}{R_1} + \frac{C_1}{C_2} \right) + R_2 C_1 s + \frac{1/RC_2}{s} \right]$

Compensator	Network	Transfer Fn
LAG		$\frac{R_2 (s + 1/R_2 C)}{R_1 + R_2 (s + \frac{1}{(R_1 + R_2)C})}$
LEAD		$\frac{s + 1/R_1 C}{s + \frac{1}{R_1 C} + \frac{1}{R_2 C}}$
LEAD-LAG		$\frac{(s + 1/R_1 C)(s + 1/R_2 C)}{s^2 + \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} \right) s + \frac{1}{R_1 R_2 C_1 C_2}}$